TrajRecovery: An Efficient Vehicle Trajectory Recovery Framework based on Urban-Scale Traffic Camera Records

Dongen Wu
Zhejiang University
Hangzhou, China
wude@zju.edu.cn

Ziquan Fang
Zhejiang University
Hangzhou, China
zqfang@zju.edu.cn

Qichen Sun
Zhejiang University
Hangzhou, China
sunqichen@zju.edu.cn

Lu Chen
Zhejiang University
Hangzhou, China
luchen@zju.edu.cn

Haiyang Hu
Huawei Cloud Computing Technologies Co., Ltd
Hangzhou, China
huhaiyang2@huawei.com

Fei Wang
Huawei Cloud Computing Technologies Co., Ltd
Hangzhou, China
wangfei1@huawei.com

Yunjun Gao
Zhejiang University
Hangzhou, China
gaoyj@zju.edu.cn

ABSTRACT

Accurate vehicle trajectory recovery enables providing indispensable data foundations in intelligent urban transportation. However, existing methods face two challenges: i) the inability to process city-wide vehicle trajectories, and ii) the dependence on a substantial amount of accurate GPS trajectories for model training, leading to poor generalization ability. To address these issues, we propose a novel trajectory recovery system based on vehicle snapshots captured by traffic cameras, named TrajRecovery. TrajRecovery consists of three main components: i) Preprocessor processes traffic cameras and vehicle snapshots to provide necessary data for trajectory recovery; ii) Spatial Transfer Probabilistic Model (STPM) integrates road conditions and driver behavior to compute turning probability at intersections; iii) Trajectory Generator utilizes the output probabilities from STPM to recover a continuous and most likely complete trajectory. We evaluate TrajRecovery on two real datasets from a city in China, demonstrating substantial performance gains compared to state-of-the-art methods. Furthermore, our system is deployed in practical applications at Huawei Company, achieving extraordinary profits in business scenarios.

CCS CONCEPTS

• **Information systems** → *Spatial-temporal systems.*

KEYWORDS

vehicle trajectory recovery, deep learning, mobile sensing

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
KDD '24, August 25–29, 2024, Barcelona, Spain
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0490-1/24/08
https://doi.org/10.1145/3637528.3671558

ACM Reference Format:
Dongen Wu, Ziquan Fang, Qichen Sun, Lu Chen, Haiyang Hu, Fei Wang, and Yunjun Gao. 2024. TrajRecovery: An Efficient Vehicle Trajectory Recovery Framework based on Urban-Scale Traffic Camera Records. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24), August 25–29, 2024, Barcelona, Spain.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3637528.3671558

1 INTRODUCTION

The proliferation of positioning devices enables the generation of vehicle trajectories, which helps in understanding traffic systems [41]. Based on the collected trajectories, researchers perform various traffic analyses such as traffic prediction [3, 19], travel time estimation [4, 37, 40], movement pattern mining [5, 6, 22, 36, 43], etc. However, in real-world applications, the raw trajectory data typically have **low sampling rates** [38]. For example, empty taxis tend to lower the sampling rates of GPS devices to save battery consumption. Complex urban environments like tall buildings or network transmission interruptions can also result in low-sampling trajectories. To provide a good-quality data foundation for trajectory analytics, **trajectory recovery** is widely studied, where the general goal is to recover accurate trajectory data based on the raw and low-sampling trajectory data.

Due to its great benefits, the data mining community has proposed many trajectory recovery methods [2, 15, 24, 30, 33], which can refer to Section 6. Specifically, they target GPS-based trajectory recovery. However, GPS devices and that-based trajectory data are only available to public transportation such as taxis and buses, rather than all vehicles [1]. Considering the relatively low installation rate of GPS devices [1, 34] and user privacy protection laws, obtaining GPS-based trajectories for most private vehicles is scarcely possible. Consequently, most of the existing studies are limited to general vehicle trajectory recovery.

In this paper, we aim to propose a new trajectory recovery method that does not rely on GPS trajectories but enables general vehicle trajectory recovery. We found that, in urban areas, traffic

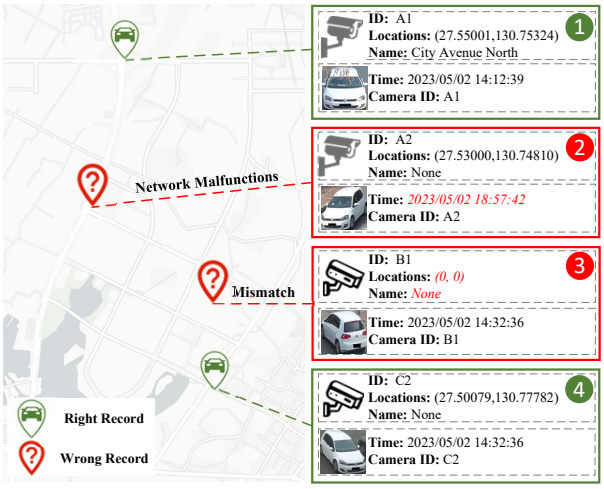


Figure 1: An Example of Sparse Vehicle Passing Records.

cameras operate 24/7, continuously capturing passing records of all vehicles [12]. Combined with vehicle re-identification technologies [8, 10, 17, 20, 26], these traffic cameras present a promising way for general vehicle recovery based on their passing records. In contrast to GPS-based trajectory recovery studies, the passing (camera) records-based methods [16, 34] involve two main tasks, i.e., vehicle re-identification and trajectory recovery. However, it is non-trivial to design such a passing record-based trajectory recovery framework due to the following three challenges.

Challenge 1: How to utilize sparse vehicle passing records for effective trajectory recovery? Although surveillance cameras operate 24/7, the observed vehicle passing records are sparse due to three aspects. i) First, not all road segments are covered by cameras [1, 28], making it challenging to predict vehicle path choices in unobserved road segments. We have collected over 20 million vehicle snapshots recorded by 491 traffic cameras in a city of China, covering a continuous period of five days. Upon analyzing these data, we find that traffic cameras only cover 7% of the road segments, leaving the remaining 93% unobservable. ii) Second, network malfunctions, image deterioration, and camera mismatch may result in the partial failure of vehicle detection, further exacerbating the sparsity problem. Fig. 1 shows an example. As observed, camera B1 failed to be accurately matched to the road network, attributed to the lack of location information and device names. Consequently, the vehicle passing records captured by these cameras were invalid. Furthermore, network malfunctions in camera A2 led to a temporal deviation in the occurrence time of the second passing record, rendering it invalid. It is important to note that, due to non-technical reasons, all coordinates and road names in this paper have undergone offsetting or obfuscation. iii) Last but not least, despite the broad distribution of traffic cameras, the distances between these devices can still be long, making it difficult for trajectory recovery. Overall, constructing an effective trajectory model using sparse vehicle passing records is challenging.

Challenge 2: How to generalize learned knowledge to unobserved road segments? Most existing GPS-trajectory-oriented recovery methods typically require massive observed trajectory data for model training. When the model encounters road segments not encompassed in the training data, it encounters difficulties in recovery. However, in real-life scenarios, a notable proportion of road segments, particularly in suburban areas, lack camera coverage. In that case, how to generalize learned knowledge from observed road segments to unobserved road segments is challenging.

Challenge 3: How to achieve personalized trajectory recovery that considers the mobility preferences of different vehicles? When performing trajectory recovery, previous methods tend to treat all drivers equally from a coarse-grained perspective and thus neglect the preference diversity in driving habits. This oversight led to the recovery of identical trajectories for different vehicles under similar contextual conditions. Therefore, we are inspired to learn the driving habits of drivers from their historical behavior, thereby achieving fine-grained trajectory recovery. Consequently, how to learn the driver's driving habits to achieve personalized trajectory recovery is challenging.

In this paper, we propose TrajRecovery, an efficient framework for vehicle trajectory recovery utilizing traffic camera records. To address ***Challenge 1***, we construct a comprehensive preprocessing pipeline to ensure the precise alignment of traffic cameras with road networks. Furthermore, we introduce a simple yet highly effective Historical Passing Record Retriever (HPRR) to retrieve comparable passing records from the historical passing records, thereby enhancing the sampling rate of sparse passing records. To address ***Challenge 2***, we design a Spatial Transfer Probability Model (STPM) to capture driver turning probability in camera-covered road segments through vehicle snapshots. To extend this knowledge to areas without camera coverage, we develop a Motion-GAT Module to capture the intrinsic connections and features between road segments by aggregating the characteristics of surrounding road segments. By incorporating the Motion-GAT Module into STPM, we ensures that STPM emphasizes the impact of road conditions on turning at the urban level, rather than specific turns between road segments, thus enabling more effective knowledge transfer. To address ***Challenge 3***, we introduce the OD-Attention Module in STPM, which incorporates attention mechanisms into the retrieval of historical driving behaviors. By utilizing driving habits, the OD-Attention Module enables vehicle-level personalized trajectory recovery, avoiding the same results for identical OD scenarios. Overall, this paper makes the following contributions.

- To the best of our knowledge, we present the first framework for trajectory recovery that solely utilizes traffic camera records instead of GPS data, offering comprehensive support for a wide range of vehicles' trajectory recovery.
- To effectively generalize learned knowledge to unobserved road segments, we develop a novel Motion-GAT Module to emphasize the road conditions on turning at the urban level, instead of specific turns between camera-covered segments.
- To enable personalized trajectory recovery, we introduce a novel OD-Attention to learn drivers' different habits from their historical driving behaviors.
- Extensive experiments using two real datasets show that TrajRecovery significantly outperforms state-of-the-art baselines, e.g., 2.6% - 12.2% improvement in recovery quality.

- Finally, the deployment of TrajRecovery in practical applications at HuaWei Company, China, showcasing outstanding performance in real-world business scenarios.

The paper is organized as follows. Section 2 presents preliminaries and a problem statement, and the proposed system is detailed in Section 3. Experimental results are reported in Section 4, and the practical deployment of TrajRecovery is introduced in Section 5. Section 6 reviews related work and Section 7 concludes this paper.

2 PRELIMINARIES

2.1 Definitions

Definition 1 (Traffic Camera). The traffic camera set is denoted as $C = \{c_1, c_2, \cdots, c_N\}$, where N is the total number of traffic cameras, and each traffic camera is represented as $c_i =< Lat, Lon, Name, Direc >$, where Lat, Lon and $Name$ represent the latitude, longitude and name of the traffic camera respectively, and $Direc$ is the shooting direction of traffic cameras (e.g., south to north). Note that, as mentioned in Challenge 1, these information is unrealiable.

Definition 2 (Road Network). The road network is an essential component for trajectory recovery. A road network is a directed graph $G = (V, E)$, where $V = \{v_1, v_2, \cdots, v_M\}$ refers to the set of vertices representing road intersections, and $E = \{e_1, e_2, \cdots, e_L\}$ refers to the set of edges representing the road segment between intersections. Each road segment $e \in E$ contains three types of properties: 1) length, width, number of lanes, number of in-degrees and out-degrees; 2) level, such as rural, provincial way, national way, etc.; 3) type, such as tunnel, pedestrian, side road, etc.

Definition 3 (Vehicle Trajectory). We define a vehicle trajectory as a sequence of time-ordered passing records $\mathcal{J}_p = [r_1, r_2, \cdots, r_n]$ for a vehicle over a continuous period. Each $r_i =< c, t, e >$ represents a passing record, indicating that vehicle p is detected by camera $r_i.c$ at time $r_i.t$ on road segment $r_i.e$, where camera $r_i.c$ is located. Further details of camera matching can be found in Section 3.1. Each passing record corresponds to a vehicle snapshot, we denote R as the set of all vehicle snapshots.

Definition 4 (OD-Pair). For each vehicle passing record r_i, the corresponding road segment $r_i.e$ is denoted as e_o, and the adjacent road segment the vehicle is approaching is designated as e_d. Notably, e_d is inferred based on the identified vehicle's lane position and turn information from snapshots and the road network G. Then, we identify the corresponding OD-pairs for each passing record, and form a OD-pair trajectory $\mathcal{J}_p^{OD} = [OD_1^{\mathcal{J}}, OD_2^{\mathcal{J}}, \cdots, OD_n^{\mathcal{J}}]$, where $OD_i^{\mathcal{J}} =< t_o, e_o, e_d >$ represents a OD-pair, indicating that vehicle p traveled from road segment $OD_i^{\mathcal{J}}.e_o$ to adjacent road segment $OD_i^{\mathcal{J}}.e_d$ at time $OD_i^{\mathcal{J}}.t_o$, where $OD_i^{\mathcal{J}}.e_o$ is $r_i.e$ and $OD_i^{\mathcal{J}}.t_o$ is $r_i.t$.

Problem Statement. Given a traffic camera set C and the vehicle snapshot data R from N traffic cameras, our task is to recover the sparse passing record trajectory into a continuous and complete trajectory based on the road network G.

3 TRAJECTORY RECOVERY SYSTEM

To address the aforementioned challenges, as shown in Fig. 2, we introduce a novel system named TrajRecovery, comprising three main components: 1) the Preprocessor, 2) Spatial Transfer Probability Model (STPM), and 3) Trajectory Generator.

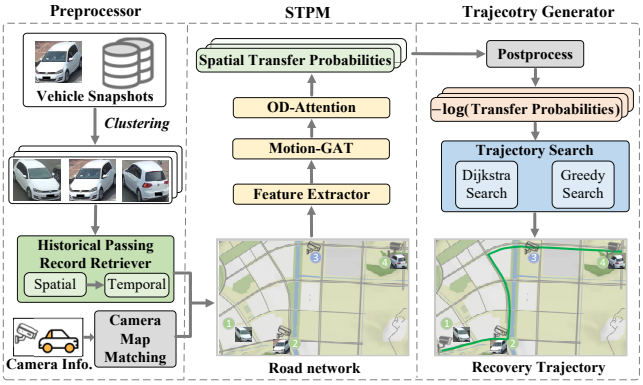


Figure 2: System architecture.

The Preprocessor consists of three steps. Initially, we match traffic cameras with road segments on the road network based on their latitude, longitude, name and direction. Drawing inspiration from previous studies [34, 35], we extract visual features and license plates from vehicles' passing snapshot records. Subsequently, we cluster vehicles based on the visual feature similarity, obtaining all passing records for each vehicle. Finally, we introduce the Historical Passing Record Retriever (HPRR) module, enhancing the sampling rate of sparse trajectories by retrieving historical passing records for the current vehicle and other vehicles.

Additionally, we devise the Spatial Transfer Probability Model (STPM) to calculate the probability of a vehicle transitioning from one road segment to its subsequent segment. The Feature Extractor generates deep representations based on road segment properties such as length, width, type, and level for each input road segment. Following that, the Motion-GAT module characterizes the traffic development and vehicle suitability of a road segment based on its relationships with surrounding segments. Considering that even with the same origin-destination (OD), different driving habits may generate distinct trajectories, we introduce the OD-Attention module to mine driving habits from a vehicle's historical driving behavior, achieving vehicle-level personalized trajectory recovery.

Finally, in the Trajectory Generator, we leverage the inference probabilities from the STPM to recovery a continuous and probabilistically optimal trajectory based on the road network.

3.1 Preprocessor

The Preprocessor encompasses traffic camera map matching, vehicle snapshot processing and clustering, and upsampling.

Traffic camera map matching. Matching traffic cameras to road segments on the road network is the foundation for trajectory recovery from vehicle passing records. Initially, we categorize all cameras into four classes based on their credibility derived from latitude, longitude, and name information. Subsequently, we employ distinct strategies to match cameras to corresponding intersections under different situations: (1) Correct location and incorrect name: match to the nearest intersection based on location; (2) Incorrect location and correct name: utilize edit distance to identify the intersection with the highest text similarity; (3) Correct location and correct name: identify the top-k nearest intersections and calculate text similarity with edit distance, selecting the intersection with the highest text similarity; (4) Incorrect location and incorrect name:

unmatchable. Once the corresponding intersection is identified, we use the camera's *Direction* to locate the road segment in the current direction of the intersection. Consequently, the camera is then associated with this road segment, represented as $c_i \rightarrow e_j$, indicating the binding of camera c_i to road segment e_j. This matching strategy successfully matches 491 out of 602 cameras, covering 96% of passing records, ensuring the availability of these records.

Vehicle snapshot processing and clustering. When a vehicle passes by, the traffic camera will capture all the content within its field of view, including noise. Consequently, preprocessing the snapshot is essential for accurate vehicle identification. Inspired by advancements in object detection algorithm [29], we first locate the vehicle and lane (e.g. left-turn lane) from the snapshot by the excellent detection ability of YOLOV7 [29], and extracting 256-dimensional vehicle appearance features f_a and vehicle plate features f_p. Inspired by [34], we then calculate the total similarity between each pair of vehicle snapshots as:

$$S_{i,j} = \frac{w_a f_a^i \cdot f_a^j + w_p f_p^i \cdot f_p^j}{w_a + w_p} \tag{1}$$

where the hyper-parameters w_a, w_p represent the weight of vehicle appearance features and plate features respectively. Considering that the plate is more recognizable than the appearance of the vehicle, we assign w_p to be larger than w_a. Then we use the total similarity of each pair of vehicle snapshots for clustering. If the total similarity between them is greater than a predefined threshold T_h, then they belong to the same cluster.

Upsampling. As mentioned in Section 1, the camera may occasionally fail, resulting in vehicles passing by without obtaining passing records. To address this, considering that people's travel often has repeatability [33], we devised a Historical Passing Record Retriever to improve the sampling rate of passing records by retrieving the vehicle's historical passing records before performing trajectory recovery. Specifically, in order to find a trajectory similar to the current trajectory from the historical passing records, we consider both spatial and temporal aspects. From a spatial perspective, for any two consecutive passing records r_i, r_{i+1} in the vehicle trajectory $\mathcal{J}_p$, we calculate the longest common subsequence (LCSS) between the subtrajectory with a context window of size k $[r_{i-k}.c, \cdots, r_i.c, r_{i+1}.c, \cdots, r_{i+1+k}.c]$ and any sub-trajectory in historical trajectory with a context window of size k. By this way, we can obtain the top-q historical sub-trajectories with the highest spatial similarity. From a temporal perspective, we sort the top-q historical subtrajectories in ascending order by $|(r_b.t - r_a.t) - (r_{i+1}.t - r_i.t)|$, where r_a, r_b represent the historical passing records of cameras $r_i.c$ and $r_{i+1}.c$ in the historical subtrajectory, and $r_b.t - r_a.t > 0$. Then we obtained the top-s subtrajectories with the highest temporal similarity from the top-q. Finally, we identify the most frequent camera c_m and average travel time t_m from c_m to $r_{i+1}.c$ between camera $r_i.c$ and camera $r_{i+1}.c$ in the subtrajectories in top-s, and insert the passing record $< c_m, r_i.t + t_m >$ between r_i and r_{i+1} to fill the missing passing record. By retrieving historical vehicle passing records, we can further increase the density of vehicle passing records, which is more conducive to subsequent trajectory recovery.

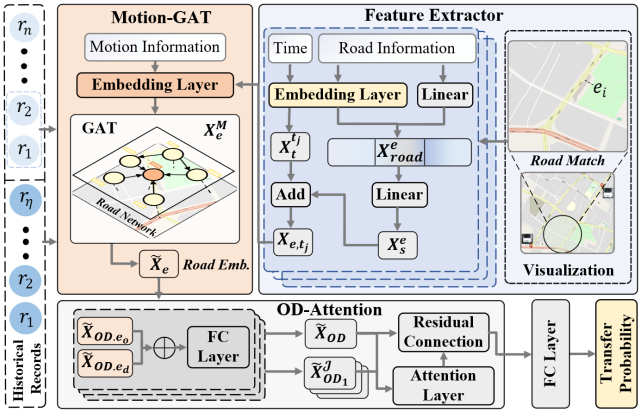


Figure 3: Spatial Transfer Probability Model.

3.2 Spatial Transfer Probability Model

To estimate the probability of a road segment transitioning to its successor, referred to as 'tuning probability', employing probability models is a widely accepted approach in trajectory recovery tasks [11, 16, 32, 35]. Considering that the traffic camera is difficult to record the continuous trajectory, previous studies rely on high sampling rate GPS trajectories to establish a transfer probability model. However, as discussed in Section 1, obtaining GPS trajectories with high sampling rates is challenging, and such data often exhibits information bias. To address it, we propose a Spatial Transfer Probability Model (STPM) to recover the most probable trajectory between consecutive passing records based on vehicle passing records.

Problem Optimization. To begin with, we refine the trajectory recovery problem into a trajectory probability calculation problem, which quantifies the probability of a certain trajectory as the recovery result, and the recovered trajectory is the trajectory with the highest probability. Traditionally, for a vehicle p, given a start passing record r_s with its road segment $r_s.e$ and an end passing record r_e with its road segment $r_e.e$ and timestamp $r_e.t$, the set of historical passing record $\mathcal{H}$, the probability of a trajectory $\mathcal{R} = \{\mathcal{R}.e_0, \cdots, \mathcal{R}.e_{|\mathcal{R}|}\}$ between r_s and r_e can be expressed as:

$$\begin{aligned} Pr(\mathcal{R}|r_s, r_e, p, \mathcal{H}) &= Pr(\mathcal{R}|r_s.e, r_e.e, r_e.t, p, \mathcal{H}) \\ &= Pr(\mathcal{R}.e_0, \cdots, \mathcal{R}.e_{|\mathcal{R}|}|r_s.e, r_e.e, r_e.t, p, \mathcal{H}) \\ &= \prod_{i=1}^{|\mathcal{R}|} Pr(\mathcal{R}.e_i|\mathcal{R}.e_0 \cdots \mathcal{R}.e_{i-1}, r_e.e, r_e.t, p, \mathcal{H}) \end{aligned} \tag{2}$$

Considering that many studies have shown that human mobility patterns follow Markov's assumptions [16, 31, 32], we can further convert formula (2) to:

$$Pr(\mathcal{R}|r_s, r_e, p, \mathcal{H}) = \prod_{i=1}^{|\mathcal{R}|} Pr(\mathcal{R}.e_i|\mathcal{R}.e_{i-1}, r_e.e, r_e.t, p, \mathcal{H}) \tag{3}$$

Accordingly, the trajectory recovery problem can be conceptualized as finding a trajectory $\mathcal{R}^*$ with the highest conditional probability:

$$\begin{aligned}\mathcal{R}^* &= \arg\max_{\forall \mathcal{R} \in G} Pr(\mathcal{R}|r_s, r_e, p, \mathcal{H}) \\ &= \arg\max_{\forall \mathcal{R} \in G} \prod_{i=1}^{|\mathcal{R}|} Pr(\mathcal{R}.e_i|\mathcal{R}.e_{i-1}, r_e.e, r_e.t, p, \mathcal{H})\end{aligned} \tag{4}$$

According to the guidance of Formula (4), we designed the STPM to calculate $Pr(\mathcal{R}.e_i|\mathcal{R}.e_{i-1}, r_s.e, r_s.t, p, \mathcal{H})$, i.e. for vehicle p, given the destination $r_e.e$, the arrival time $r_e.t$, and the set of historical passing record $\mathcal{H}$, we aims to calculate the probability of any road segment $\mathcal{R}.e_{i-1}$ of the road network to its successor road segment $\mathcal{R}.e_i$. For each $\mathcal{R} \in G$, we generate the corresponding OD-pair trajectory $\mathcal{R}^{OD} = [OD_1^{\mathcal{R}}, OD_2^{\mathcal{R}}, \cdots, OD_{|\mathcal{R}|}^{\mathcal{R}}]$, where $OD_i^{\mathcal{R}} =< r_e.t, \mathcal{R}.e_{i-1}, \mathcal{R}.e_i >, \forall i, 1 \le i \le |\mathcal{R}|$. Since $\mathcal{R}$ can represent any trajectory from origin road segment $r_s.e$ to destination road segment $r_e.e$, we can also denote $OD_i^{\mathcal{R}} =< r_e.t, e_a, e_b >$, where e_b is the successor road segment of road segment e_a . Therefore, we can simplify $OD_i^{\mathcal{R}}$ as OD_{ab}.

3.2.1 Feature Extractor. To obtain dense representations of each road segment in the road network, we devise a road segment feature extractor, as show in Fig. 3. First, to represent whether the road segment is suitable for vehicle traffic and the level of road development, we jointly embed the road segment's properties and time into dense representations as the input of following modules. Specifically, for properties of category types (e.g., type, level), we set up two trainable embedding matrix $Emb_{type} \in \mathbb{R}^{|\#types| \times d}$ and $Emb_{level} \in \mathbb{R}^{|\#levels| \times d}$, therefore, the road type embedding vector and road level embedding vector of road segment e_i can be represented as $X_{type}^{e_i} \in \mathbb{R}^d$ and $X_{level}^{e_i} \in \mathbb{R}^d$ respectively. And for numerical types (e.g., length, width, the distance close to the destination), we use linear transformation to obtain d dimension vectors $X_{num}^{e_i} \in \mathbb{R}^d$. We concatenate the above vectors to obtain road segment-level hidden-state vector $X_{road}^{e_i} \in \mathbb{R}^{3d}$, followed by linear transformation to obtain static representation of road segment $X_s^{e_i} \in \mathbb{R}^d$. Besides, consider that the traffic conditions on road segments may vary during different time periods (such as morning and evening rush hours), we divide a day into 48 time periods and set up a trainable embedding matrix $Emb_{time} \in \mathbb{R}^{48 \times d}$. Finally, the spatio-temporal representation of road segment e_i at time t_j can be represented as:

$$X_{e_i,t_j} = X_s^{e_i} + X_t^{t_j} \tag{5}$$

3.2.2 Motion-GAT. The previously extracted road segment spatial-temporal features focus primarily on the individual road segments. However, it is insufficient for effectively generalize learned knowledge from camera-covered segments to unobserved road segments. To this end, considering that the traffic development level of the road segment is influenced by its adjacent road segments, we propose the Motion-GAT module. This module first extracts the subgraph related to road segments from the road network G and then uses the self-attention mechanism to capture the interaction information between adjacent road segments and the current road segment.

Firstly, we extract subgraphs for each road segment e_i from the road network. For each road segment e_i, its adjacent road segment is denoted as $e_a \in \overline{Adj_i}$, $Adj_i = \{\overline{Adj_i}, e_i\}$. We represented the motion relationship between e_a and e_i by concatenating their turn information (e.g., left turn, right turn, straight, U-turn) and transfer relationship (e.g., $e_i \rightarrow e_a$ or $e_a \rightarrow e_i$), resulting 8 distinct motion relationships. Subsequently, We set a trainable d-dimensional motion vector $Emb_m \in \mathbb{R}^{8 \times d}$ for each motion relationship. Thus, the motion vector for e_a is denoted as $M_{e_a,t_j} \in \mathbb{R}^d$. For clarity of formulas, we omit the time variable t_j in the following discussion. Moreover, we add the motion vector to the spatial-temporal feature and obtain the final road segment representation $X_{e_a}^M$:

$$X_{e_a}^M = X_{e_a} + M_{e_a} \tag{6}$$

where X_{e_a} represents the spatio-temporal feature of road segment e_a and M_{e_a} denotes the motion vector of e_a. This approach further strengthens the relationship between e_a and e_i through the motion vector. Then we use a multi-head self-attention mechanism to focus on different motions and the impact of different adjacent road segments on the current road segment from multiple perspectives.

Specifically, we define the feature correlation between the adjacent road segment e_a that has been fused with the motion vector and the current road segment e_i under the h-head attention as:

$$\begin{aligned}\alpha^{(h)}(e_i, e_a) &= \frac{exp(\psi(X_{e_i}, X_{e_a}^M))}{\sum_{a'=1}^{|Adj_i|} exp(\psi(X_{e_i}, X_{e_{a'}}^M))} \\ \psi(X_{e_i}, X_{e_a}^M) &= \frac{< W_Q^{(h)} X_{e_i}, W_K^{(h)} X_{e_a}^M >}{\sqrt{d'}}\end{aligned} \tag{7}$$

where $W_Q^{(h)}, W_K^{(h)} \in \mathbb{R}^{d' \times d}$ are learnable parameters, $<,>$ is the inner product function, and $d' = d/H$. Next, we generate the aggregated representation of road segment e_i by correlation $\alpha^{(h)}(e_i, e_a)$:

$$\begin{aligned}\tilde{X}_{e_i} &= \tilde{X}_{e_i}^{(1)} \oplus \tilde{X}_{e_i}^{(2)} \oplus \cdots \oplus \tilde{X}_{e_i}^{(H)} \\ \tilde{X}_{e_i}^h &= \sum_{a'=1}^{|Adj_i|} \alpha^{(h)}(e_i, e_{a'})(W_V^{(h)} X_{e_{a'}}^M)\end{aligned} \tag{8}$$

where $W_V^{(h)} \in \mathbb{R}^{d' \times d}$ is learnable parameters, and $\oplus$ is the concatenation operator, and H is the number of total heads. As such, we can integrate the structural information of the road network into the representation of road segments, and in the same way, we can obtain the representation of each road segment in the road network.

3.2.3 OD-Attention. As mentioned in Challenge 3, different vehicles may produce different driving trajectories even at the same time and with the same OD. Thus, we develop OD-Attention to extract the driving habits of drivers.

Assuming that the vehicle p is currently located in road segment e_a, where e_a represents a road segment in a trajectory between passing record r_s and r_e. The first step is to retrieval the historical OD-pairs before time $r_e.t$, and obtain the representation for each OD-pair. Specifically, for OD-pair $OD_{ab} \in \mathcal{R}^{OD}$, its d_{OD} dimension representation $\tilde{X}_{OD_{ab}} \in \mathbb{R}^{d_{OD}}$ can be calculated as:

$$\tilde{X}_{OD_{ab}} = \mathcal{M}(\tilde{X}_{OD_{ab}.e_o} \oplus \tilde{X}_{OD_{ab}.e_d}) \tag{9}$$

where $\mathcal{M}$ denote two-layer fully-connected neural network function. As such, we further obtain the representation sequence of vehicle OD-pair trajectory $\mathcal{J}_p^{OD}$ as $\tilde{X}_{OD^{\mathcal{J}}} = [\tilde{X}_{OD_1^{\mathcal{J}}}, \tilde{X}_{OD_2^{\mathcal{J}}}, \cdots, \tilde{X}_{OD_\eta^{\mathcal{J}}}]$, where η is the number of total historical OD-pairs. The next step is to extract the driving habits among historical OD-pairs. Intuitively, we believe that driving habits can be seen as similar choices that drivers habitually make under similar road conditions. Therefore, we calculate the similarity between the OD-pair representation $\tilde{X}_{OD_{ab}}$ of the current OD-pair OD_{ab} and the OD-pair representations $\tilde{X}_{OD_j^{\mathcal{J}}}$ of the historical OD-pairs, and then selectively incorporate the historical OD-pair features $\tilde{X}_{OD_j^{\mathcal{J}}}$ into the current OD-pair features $\tilde{X}_{OD_{ab}}$ by multi-head attention mechanism:

$$\begin{aligned} \tilde{X}_{OD_{ab}}^{\mathcal{A}} &= \tilde{X}_{OD_{ab}}^{\mathcal{A}(1)} \oplus \tilde{X}_{OD_{ab}}^{\mathcal{A}(2)} \oplus \cdots \oplus \tilde{X}_{OD_{ab}}^{\mathcal{A}(H)} \\ \tilde{X}_{OD_{ab}}^{\mathcal{A}(h)} &= \sum_{j'=1}^{n} \alpha^{(h)}(OD_{ab}, OD_j^{\mathcal{J}})(W_{V'}^{(h)} \tilde{X}_{OD_{j'}^{\mathcal{J}}}) \end{aligned} \tag{10}$$

$$\begin{aligned} \alpha^{(h)}(OD_{ab}, OD_j^{\mathcal{J}}) &= \frac{exp(\psi(\tilde{X}_{OD_{ab}}, \tilde{X}_{OD_j^{\mathcal{J}}}))}{\sum_{j'=1}^{n} exp(\psi(\tilde{X}_{OD_i^{\mathcal{R}}}, \tilde{X}_{OD_{j'}^{\mathcal{J}}}))} \\ \psi(\tilde{X}_{OD_{ab}}, \tilde{X}_{OD_j^{\mathcal{J}}}) &= \frac{< W_{Q'}^{(h)} \tilde{X}_{OD_{ab}}, W_{K'}^{(h)} \tilde{X}_{OD_j^{\mathcal{J}}} >}{\sqrt{d'_{OD}}} \end{aligned} \tag{11}$$

where $W_{Q'}^{(h)}, W_{K'}^{(h)}, W_{V'}^{(h)} \in \mathbb{R}^{d'_{OD} \times d_{OD}}$ are learnable parameters, and $d'_{OD} = d_{OD}/H$. To preserve the representation of the original OD-pair, we further use standard residual connections:

$$\overline{X}_{OD_{ab}} = \sigma(\tilde{X}_{OD_{ab}}^{\mathcal{A}} + W^r \tilde{X}_{OD_{ab}}) \tag{12}$$

where $W^r \in \mathbb{R}^{d_{OD} \times d_{OD}}$ is learnable projection matrix, and σ is non-linear activation function e.g. *ReLU* function.

With the OD-attention layer, the representation of OD-pair $\tilde{X}_{OD_{ab}}$ can be updated into a more informative representation $\overline{X}_{OD_{ab}}$, incorporating the driving habit features from historical passing records.

3.2.4 Transfer Probabilistic Generator. Assuming that the vehicle is currently located in road segment e_a, it can then travel to any adjacent road segment $e_b \in Adj_a$. Thus, we obtain the transfer probability of $e_a \rightarrow e_b$ by:

$$Pr(e_b|e_a, r_e.e, r_e.t, p, \mathcal{H}) = \frac{exp(\mathcal{F}(\overline{X}_{OD_{ab}}))}{\sum_{b'=1}^{|Adj_a|} exp(\mathcal{F}(\overline{X}_{OD_{ab'}}))} \tag{13}$$

where $\mathcal{F}$ denote two-layer fully-connected neural network function:

$$\mathcal{F}(X) = \sigma(XW_1 + b_1)W_2 \tag{14}$$

Here, $W_1 \in \mathbb{R}^{d_{OD} \times d_1}$ and $W_2 \in \mathbb{R}^{d_1 \times 1}$ are two learnable weights, and b_1 is the learnable bias.

3.3 Trajectory Generator

Given the vehicle trajectory $\mathcal{J}_p$ of vehicle p, based on formula (4) and the transfer probability output of each road segment $e_a \in E$ transfer to its adjacency road segment $e_b \in Adj_a$ by the STPM. We sequentially recover the trajectory between passing record $r_i \in \mathcal{J}_p$ and passing record $r_{i+1} \in \mathcal{J}_p$. Inspired by [11], we simplify the Formula 4 as:

$$\begin{aligned} \mathcal{R}^* &= \arg \max_{\forall \mathcal{R} \in G} \prod_{i=1}^{|\mathcal{R}|} Pr(\mathcal{R}.e_i|\mathcal{R}.e_{i-1}, r_e.e, r_e.t, p, \mathcal{H}) \\ &= \arg \min_{\forall \mathcal{R} \in G} \sum_{i=1}^{|\mathcal{R}|} -log(Pr(\mathcal{R}.e_i|\mathcal{R}.e_{i-1}, r_e.e, r_e.t, p, \mathcal{H})) \end{aligned} \tag{15}$$

Based on that, we can form a road network graph with road segments as nodes and the connection relationships between road segments as edges based on the connection relationship of road segments in the road network G. We can regard the transfer probability as the edge weight between nodes, i.e. road segments, so we can transform the problem into the shortest path finding problem and use Dijkstra's Algorithm to solve it.

3.4 Training

Finally, we elaborate on the end-to-end training process of STPM. As mentioned in Section 1, we obtain snapshots of vehicles captured by traffic cameras, extracting lane information and corresponding turning details. Therefore, we construct training samples using snapshots with explicit turning information (detailed construction rules are introduced in Appendix A). Each training sample is represented as (x_i, y_i), where $x_i =< e_a, e_b, r_e, p, \mathcal{H}_p >$ signifies that the vehicle p is currently on road segment e_a, and given its destination at road segment $r_e.e$ and arrive time $r_e.t$, the probability of heading towards adjacent road segment $e_b \in Adj_a$ is y. In a mini-batch of b training samples, the training objective is to minimize the model's mean squared error using the following loss function:

$$L(\theta) = \frac{1}{b} \sum_{i=1}^{b} (y_i - f(x_i, \theta))^2 + \lambda ||\theta||^2 \tag{16}$$

Here, $f(x, \theta)$ represents the output of STPM, θ denotes all trainable parameters in the model, and λ is a parameter controlling the regularization strength. Our goal is to train the model and find the optimal parameters θ that minimize the loss function $L(\theta)$.

4 EXPERIMENTS

4.1 Experimental Settings

4.1.1 Datasets. **Camera Record Data.** To demonstrate the robustness of TrajRecovery, we use two different-scale real traffic camera record datasets, which are collected from a region of a city in China, which encompasses 975 traffic cameras, 5,597 road intersections, and 13,579 road segments. Specifically, the larger dataset is D5, which comprises over 20 million vehicle snapshots over 5 days. The smaller dataset is D1, which consists of 1.5 million vehicle snapshots during the morning rush hour on the workday.

Ground Truth Data. To evaluate the recovery performance, we require ground truth data. Corresponding to the time cover range of D5 and D1, our volunteers collect 515 and 196 real GPS trajectories, respectively, while driving on the roads. Then, we perform map-matching [21] to align these GPS trajectories with the road network, which helps evaluate trajectory recovery quality.

4.1.2 Baselines. We compare TrajRecovery with six baselines.

Table 1: Overall Performance Comparison.

Method	D1				D5			
	LCSS	EDR	STLC	DTW	LCSS	EDR	STLC	DTW
SP	0.4716±0.0031	30.5164±0.1059	0.6128±0.0058	1253±11	0.5831±0.0052	36.1543±0.1129	0.5043±0.0081	1863±30
VeTrac	0.3921±0.0108	18.9641±0.6018	0.7681±0.0130	896±38	0.5031±0.0126	23.5163±0.7158	0.5968±0.0146	1653±53
MMVC	0.3761±0.0124	16.5176±0.7426	0.8214±0.0105	841±40	0.4713±0.0135	19.9643±0.8201	0.6514±0.0117	1568±51
CFST	0.3495±0.0078	15.1536±0.5971	0.8537±0.0126	726±28	0.4482±0.0103	18.7694±0.6391	0.6786±0.0138	1496±42
CFST-C	0.3351±0.0071	14.9658±0.5820	0.8602±0.0107	719±29	0.4324±0.0098	18.0659±0.6018	0.6812±0.0118	1406±38
TrajRecovery-G	0.3356±0.0116	15.1443±0.6419	0.8616±0.0130	728±36	0.4218±0.0120	17.5011±0.6310	0.6943±0.0107	1367±28
TrajRecovery	**0.3179±0.0082**	**13.8169±0.5981**	**0.8826±0.0118**	**653±31**	**0.4096±0.0108**	**16.6115±0.6189**	**0.7281±0.0116**	**1235±47**
Gain	5.1%	7.7%	2.6%	9.2%	5.3%	8.1%	6.9%	12.2%

- Shortest Path (SP) [39]. Initially, we use the vehicle reidentification in the Preprocessor module to cluster vehicles. Then, we employ this method to generate the shortest path.
- VeTrac [28]. It utilizes the graph convolutional process to maintain identity consistency across different camera observations. Due to memory limitations, following the previous study [34], we employ the k-means clustering.
- MMVC [16]. It jointly optimizes vehicle reidentification and trajectory recovery. It clusters vehicles using visual features and employs the Hidden Markov Model (HMM) for trajectory recovery based on vehicle clusters.
- CFST [34]. Similar to MMVC [16], it clusters vehicles using visual features and employs GPS trajectories to train a uniform Dirichlet prior model and a temporal model. Specifically, we utilize 84,153 trajectories recorded over 24 hours on a day not included in the dataset for model training.
- CFST-C [35]. This is an optimized version of CFST [34], dividing the entire map area into multiple blocks and recovering and concatenating trajectories blockwise.
- TrajRecovery-G. We replace the Dijkstra algorithm used in the trajectory generator with a greedy algorithm. It selects the road segment with the highest probability at each step during the trajectory generation process.

4.1.3 Evaluation Metrics. Referring to previous studies, we utilize four metrics to evaluate the precision of trajectory recovery, i.e., Longest Common SubSequence (LCSS), Edit Distance on Real sequence (EDR), Spatio-Temporal Linear Combine distance (STLC) [27], and Dynamic Time Warping (DTW) [25]. The larger the STLC value, the better the quality of trajectory recovery; the smaller the LCSS/EDR/DTW value, the better the quality of trajectory recovery.

4.1.4 Parameter Settings. All experiments were conducted on a Linux server equipped with PyTorch 1.10.1, 2 NVIDIA TESLA T4 GPUs, and 64GB memory. These programs were implemented using Python. In the Preprocessor module, we set the similarity threshold $T_h = 0.8$, the context window $k = 2$, $q = 50$, and $s = 20$. In the Feature Extractor module, we configure the dimension d of each dense representation to be 128. In the OD-Attention module, we set the dimension d_{OD} to 64. In multi-head attention, we employ 4 heads to generate attention from different perspectives. We use the Adam optimizer [14] for training, where the initial learning rate is 0.0001 and the mini-batch size is 128. The size of training/validation/testing is set to 0.8/0.1/0.1 for STPM, and select the model that performs best on the testing set for trajectory recovery. The Xavier initializer [7] is employed to initialize the model parameters, and the associated parameters adhere to the configurations outlined in the original paper. We present the replication results for all baselines, using hyperparameters consistent with those specified in the original paper.

4.2 Overall Performance Comparison

Table 1 shows the overall performance of all methods on two datasets, where the best result in each metric (column) is shown in bold. Overall, we yield the following observations.

The first observation is that our model consistently outperforms baselines on all four metrics for both datasets, showing great superiority, e.g., up to 12% improvement. Second, SP performs poorly, as individuals may consider various factors when selecting routes while driving. Even in similar road conditions, drivers may choose longer routes due to driving habits. In contrast, our methods comprehensively consider factors such as road conditions and driving habits, making them more aligned with real-world scenarios. Another interesting observation is both VeTrac and MMVC require a high density of traffic cameras to provide sufficient information. This leads to their suboptimal performance in suburban areas with lower traffic camera density. Although CFST and CFST-C perform better than VeTrac and MMVC by utilizing GPS trajectories from other vehicles in their trajectory recovery, biases in path preferences may occur, causing the model to fail on road segments not covered by GPS trajectories. In contrast, our methods employ a universal method for trajectory recovery based on road conditions and individual driving habits, even in areas without camera coverage. We also observe that in the D1 dataset, CFST-C is comparable to the TrajRecovery-G version. This is because, during the morning rush hour, people's travel patterns tend to be relatively fixed, resulting in concentrated driving trajectories. The final observation is that, when replacing the Dijkstra algorithm with the greedy algorithm in TrajRecovery, its performance declined. This is mainly because consistently moving to the road segment with the highest probability can result in local optima rather than global optima in certain situations. It may also lead to the occurrence of trajectory "loops," where the road segment with the highest probability forms a loop, preventing the trajectory from reaching the destination.

Table 2: Impact of each Component (OD: OD-Attention, M: Motion-GAT, HPRR: Historical Passing Record Retriever).

Ablation	LCSS	EDR	STLC	DTW
-OD & M	0.4367±0.0128	17.9584±0.6241	0.6894±0.0125	1437±56
-M	0.4208±0.0110	17.9123±0.6110	0.6953±0.0120	1342±53
-HPRR	0.4195±0.0107	17.7614±0.5917	0.6927±0.0102	1415±43
-OD	0.4164±0.0113	17.6615±0.6071	0.6913±0.0127	1361±50
All	**0.4096±0.0108**	**16.6115±0.6189**	**0.7281±0.0116**	**1235±47**

4.3 Ablation Study

We study each module to influence the final results. To achieve this, we systematically performed ablation experiments by sequentially removing each module. The results are presented in Table 2, and for a more comprehensive conclusion, we conducted the analysis using the information-rich D5 dataset. It is evident from the results that the performance of the complete version of the system is optimal, indicating that each module contributes to the overall system performance. The most notable performance decline occurred when the Motion-GAT module was ablated. This is attributed to the significance of factors such as the developmental level and traffic conditions of road segments, crucial elements influencing a driver's decision to choose a particular route. Additionally, the traffic conditions of surrounding road segments also impact the driver's decision on whether to select a particular route.

4.4 Robustness Study

We conducted experiments to assess the robustness of TrajRecovery in various scenarios. As mentioned in the first chapter, traffic cameras may occasionally malfunction, resulting in instances where vehicles passing through are not captured. Specifically, we investigated the system's recovery performance under different percentages of missing vehicle records. For a particular vehicle's complete record, we retained the first and last entries, randomly removing varying percentages of intermediate entries from the vehicle record database. The results are presented in Fig. 4, indicating that as the percentage of missing entries increases, trajectory recovery errors gradually escalate. However, our model consistently outperforms the comparison methods in terms of performance.

4.5 Parameter Study

We investigate the sensitivity of three parameters: vehicle snapshot similarity threshold T_h, the road segment representation dimension d, and the head number H. The results are shown in Appendix B.

4.6 Case Study

We selected a representative trajectory from the test trajectories to illustrate the effectiveness of TrajRecovery. As shown in Fig. 5, compared to CFST-C, the results of TraRecovery are closer to the ground truth. Specifically, we first identified six passing records of the vehicle through vehicle re-identification. Subsequently, using the Historical Passing Record Retriever, we retrieved the historical passing records of the vehicle. The retrievals indicated that the vehicle passed through the camera at the red coordinate point

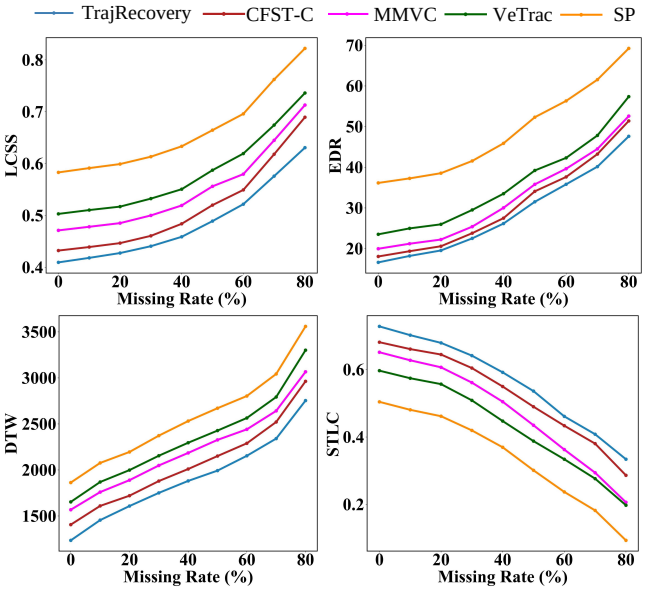


Figure 4: The influence of the missing rate.

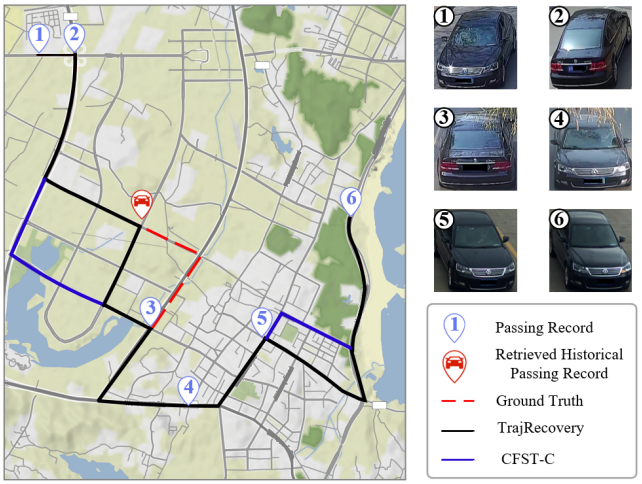


Figure 5: Visualization of trajectory recovery by TrajRecovery and CFST-C.

under similar contextual conditions multiple times. This preliminary increase in the sampling rate of passing record trajectories before trajectory recovery helps reduce uncertainty. Moreover, at the fifth passing record, the vehicle chose to turn left instead of going straight, which differs from the choice of the majority. This decision was influenced by the driver's historical OD-pair preferences, indicating a preference for roads with better conditions rather than the closest ones. However, CFST-C relies heavily on a large volume of GPS trajectories for trajectory recovery, making it prone to failure when encountering intersections not encountered during the training phase. In contrast, TrajRecovery, even in road segments not covered by traffic cameras, can make road selections based on road conditions and driver habits.

5 PRACTICAL DEPLOYMENT

Based on the trajectory recovery results provided by TrajRecovery, we have developed a web-based system at Huawei Cloud. This system leverages recovered trajectories for various downstream applications. For instance, Fig. 6 showcases the traffic flow monitoring interface, allowing administrators to interact with the system in the following scenarios. (i) Traffic Volume Visualization. Utilizing a heat map displayed in the upper left corner of Fig. 6, we illustrate the traffic volume at each intersection. Color variations depict the congestion or smooth flow of traffic conditions. (ii) Crowded Intersection Ranking. As depicted in the lower right part of Fig. 6, it provides periodic traffic statistics, presenting the number of vehicles during different periods and at specific intersections (e.g., Road 22). (iii) Traffic Volume Statistics. By selecting a specific region, a traffic volume trend-line for the past 24 hours is plotted in the lower right part of Fig. 6. Furthermore, administrators can easily obtain precise volume information by hovering the mouse over a specific time interval. Additionally, we have successfully deployed TrajRecovery in a city in China for traffic signal control, bringing in a noteworthy 12.5% increase in the average speed of intersections.

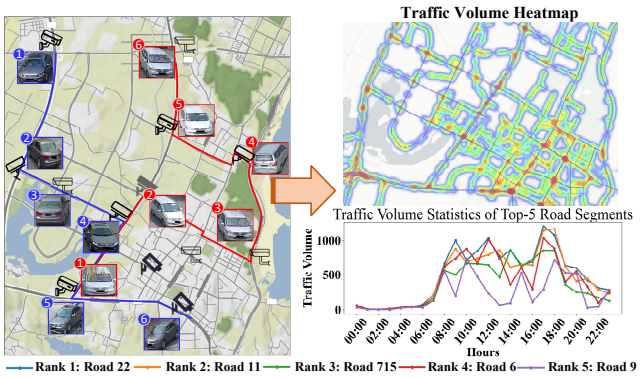


Figure 6: TrajRecovery deployment at HuaWei Cloud.

6 RELATED WORK

6.1 Trajectory Recovery

Depending on the data source of sparse vehicle trajectories, it can be divided into two categories.

The first category primarily focuses on sparse GPS trajectories [15, 24, 30, 33], which are not the scope of this paper.

The second category takes sparse snapshots captured by traffic cameras as input, which often requires vehicle re-identification processing, facing missed identifications, misidentifications, and aggravated sparsity. Cao et al. [1] utilized high-sampling-rate GPS trajectories from commercial vehicles to assist in recovering low-sampling-rate trajectories. Tong et al. [28] fused vehicle movement correlations and visual features to reduce uncertainties in identifying vehicles, requiring a higher camera distribution rate to enhance trajectory recovery accuracy. Qi et al. [23] performed trajectory recovery on automatic license plate recognition data obtained from cameras. Lin et al. [16] recognized the correlation between vehicle re-identification and trajectory recovery, proposing a joint optimization framework for both tasks. Yu et al. [34] optimized previous efforts by combining vehicle visual features and dynamic spatio-temporal constraint features for vehicle re-identification. Recently, Yu et al. [35] divided the entire city into multiple regions, performed trajectory recovery within each region, and then concatenated the complete trajectories. However, these approaches require a substantial amount of GPS trajectories for training trajectory recovery models and may not perform well in areas not covered by cameras.

6.2 Vehicle Re-identification

Vehicle re-identification involves clustering the same vehicle from a large number of snapshots captured by cameras, serving as an essential foundation for trajectory recovery. Numerous CNN-based methods have been proposed. Liu et al. [18] introduced a coarse-to-fine vehicle retrieval framework that utilizes multi-modal features, such as visual features, license plate information, camera location, and other contextual details. Zhou et al. [42] employed reverse learning and a viewpoint-aware attention model to acquire viewpoint-aware representations for vehicle re-identification. [13] proposed AAVER, a dual-path model with adaptive attention for vehicle re-identification, which uses key part features of vehicles to distinguish local features in different regions and concentrates attention on the most informative key points to capture local discriminative features. Shen et al. [26] introduced a Graph Interactive Transformer framework that utilizes the interaction between graphs and transformers to effectively coordinate local and global features. Meng et al. [20] extracted features from different parts of vehicles and used commonly visible attention to calculate the similarity between different parts. He et al. [9] proposed a simple yet effective method of part-regularized discriminative feature preservation, focusing attention on vehicle-specific features to enhance sensitivity to subtle differences.

7 CONCLUSION

In this paper, we propose the TrajRecovery framework for recovering urban-scale vehicle trajectories using vehicle snapshots captured by traffic cameras. TrajRecovery generalizes knowledge learned from camera-covered road segments to uncovered road segments, and learns driver-specific driving habits based on historical driving behavior, enabling vehicle-level personalized trajectory recovery. Additionally, we develop a web-based system TrajRecovery at HuaWei Cloud, which leverages recovered trajectories for various downstream applications. Experiments on two real-world datasets validate TrajRecovery's effectiveness and utility improvement on the business scenarios.

8 ACKNOWLEDGMENTS

This work was supported in part by the NSFC under Grants No. (62025206, U23A20296, and 62102351), the Zhejiang Province's "Lingyan" R&D Project under Grant No 2024C01259, the Ningbo Science and Technology Special Innovation Projects under Grants No. (2022Z095 and 2021Z019), and the Hangzhou Technology Bureau AI Research Project under Grant No. 2022AIZD0116. Yunjun Gao is the corresponding author of the work.

REFERENCES

[1] Zijian Cao, Dong Zhao, Hanxing Song, Haitao Yuan, Qiyue Wang, Huadong Ma, Jianjun Tong, and Chang Tan. 2023. F 3 VeTrac: Enabling Fine-grained, Fully-road-covered, and Fully-individual penetrative Vehicle Trajectory Recovery. *IEEE Transactions on Mobile Computing* (2023).

[2] Yuqi Chen, Hanyuan Zhang, Weiwei Sun, and Baihua Zheng. 2023. Rntrajrec: Road network enhanced trajectory recovery with spatial-temporal transformer. In *2023 IEEE 39th International Conference on Data Engineering (ICDE)*. IEEE, 829–842.

[3] Ziquan Fang, Dongen Wu, Lu Pan, et al. 2022. When transfer learning meets cross-city urban flow prediction: spatio-temporal adaptation matters. *IJCAI'22* (2022), 2030–2036.

[4] Kun Fu, Fanlin Meng, Jieping Ye, and Zheng Wang. 2020. CompactETA: A fast inference system for travel time prediction. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 3337–3345.

[5] Qiang Gao, Fan Zhou, Kunpeng Zhang, Goce Trajcevski, Xucheng Luo, and Fengli Zhang. 2017. Identifying Human Mobility via Trajectory Embeddings.. In *IJCAI*, Vol. 17. 1689–1695.

[6] Fosca Giannotti, Mirco Nanni, Dino Pedreschi, Fabio Pinelli, Chiara Renso, Salvatore Rinzivillo, and Roberto Trasarti. 2011. Unveiling the complexity of human mobility by querying and mining massive trajectory data. *The VLDB Journal* 20 (2011), 695–719.

[7] Xavier Glorot and Yoshua Bengio. 2010. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 249–256.

[8] Haiyun Guo, Chaoyang Zhao, Zhiwei Liu, Jinqiao Wang, and Hanqing Lu. 2018. Learning coarse-to-fine structured feature embedding for vehicle re-identification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 32.

[9] Bing He, Jia Li, Yifan Zhao, and Yonghong Tian. 2019. Part-regularized near-duplicate vehicle re-identification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3997–4005.

[10] Mingdi Hu, Long Bai, Jiulun Fan, Sirui Zhao, and Enhong Chen. 2023. Vehicle color recognition based on smooth modulation neural network with multi-scale feature fusion. *Frontiers of Computer Science* 17, 3 (2023), 173321.

[11] Jayant Jain, Vrittika Bagadia, Sahil Manchanda, and Sayan Ranu. 2021. NeuroMLR: Robust & reliable route recommendation on road networks. *Advances in Neural Information Processing Systems* 34 (2021), 22070–22082.

[12] Sultan Daud Khan and Habib Ullah. 2019. A survey of advances in vision-based vehicle re-identification. *Computer Vision and Image Understanding* 182 (2019), 50–63.

[13] Pirazh Khorramshahi, Amit Kumar, Neehar Peri, Sai Saketh Rambhatla, Jun-Cheng Chen, and Rama Chellappa. 2019. A dual-path model with adaptive attention for vehicle re-identification. In *Proceedings of the IEEE/CVF international conference on computer vision*. 6132–6141.

[14] Diederik P Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).

[15] Dedong Li, Ziyue Li, Zhishuai Li, Lei Bai, Qingyuan Gong, Lijun Sun, Wolfgang Ketter, and Rui Zhao. 2023. A Critical Perceptual Pre-trained Model for Complex Trajectory Recovery. In *Proceedings of the 2nd ACM SIGSPATIAL International Workshop on Searching and Mining Large Collections of Geospatial Data*. 1–8.

[16] Zongyu Lin, Guozhen Zhang, Zhiqun He, Jie Feng, Wei Wu, and Yong Li. 2021. Vehicle Trajectory Recovery on Road Network Based on Traffic Camera Video Data. In *Proceedings of the 29th International Conference on Advances in Geographic Information Systems*. 389–398.

[17] Xinchen Liu, Wu Liu, Huadong Ma, and Huiyuan Fu. 2016. Large-scale vehicle re-identification in urban surveillance videos. In *2016 IEEE international conference on multimedia and expo (ICME)*. IEEE, 1–6.

[18] Xinchen Liu, Wu Liu, Tao Mei, and Huadong Ma. 2016. A deep learning-based approach to progressive vehicle re-identification for urban surveillance. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11-14, 2016, Proceedings, Part II 14*. Springer, 869–884.

[19] Yisheng Lv, Yanjie Duan, Wenwen Kang, Zhengxi Li, and Fei-Yue Wang. 2014. Traffic flow prediction with big data: A deep learning approach. *IEEE Transactions on Intelligent Transportation Systems* 16, 2 (2014), 865–873.

[20] Dechao Meng, Liang Li, Xuejing Liu, Yadong Li, Shijie Yang, Zheng-Jun Zha, Xingyu Gao, Shuhui Wang, and Qingming Huang. 2020. Parsing-based view-aware embedding network for vehicle re-identification. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 7103–7112.

[21] Paul Newson and John Krumm. 2009. Hidden Markov map matching through noise and sparseness. In *Proceedings of the 17th ACM SIGSPATIAL international conference on advances in geographic information systems*. 336–343.

[22] Xiao Pan, Lei Wu, Fenjie Long, and Ang Ma. 2022. Exploiting user behavior learning for personalized trajectory recommendations. *Frontiers of Computer Science* 16 (2022), 1–12.

[23] Xinyi Qi, Yanjie Ji, Wenhao Li, and Shuichao Zhang. 2021. Vehicle trajectory reconstruction on urban traffic network using automatic license plate recognition data. *IEEE Access* 9 (2021), 49110–49120.

[24] Huimin Ren, Sijie Ruan, Yanhua Li, Jie Bao, Chuishi Meng, Ruiyuan Li, and Yu Zheng. 2021. Mtrajrec: Map-constrained trajectory recovery via seq2seq multi-task learning. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. 1410–1419.

[25] Stan Salvador and Philip Chan. 2007. Toward accurate dynamic time warping in linear time and space. *Intelligent Data Analysis* 11, 5 (2007), 561–580.

[26] Fei Shen, Yi Xie, Jianqing Zhu, Xiaobin Zhu, and Huanqiang Zeng. 2023. Git: Graph interactive transformer for vehicle re-identification. *IEEE Transactions on Image Processing* 32 (2023), 1039–1051.

[27] Han Su, Shuncheng Liu, Bolong Zheng, Xiaofang Zhou, and Kai Zheng. 2020. A survey of trajectory distance measures and performance evaluation. *The VLDB Journal* 29 (2020), 3–32.

[28] Panrong Tong, Mingqian Li, Mo Li, Jianqiang Huang, and Xiansheng Hua. 2021. Large-scale vehicle trajectory reconstruction with camera sensing network. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*. 188–200.

[29] Chien-Yao Wang, Alexey Bochkovskiy, and Hong-Yuan Mark Liao. 2023. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7464–7475.

[30] Jingyuan Wang, Ning Wu, Xinxi Lu, Wayne Xin Zhao, and Kai Feng. 2019. Deep trajectory recovery with fine-grained calibration using kalman filter. *IEEE Transactions on Knowledge and Data Engineering* 33, 3 (2019), 921–934.

[31] Jingyuan Wang, Ning Wu, Wayne Xin Zhao, Fanzhang Peng, and Xin Lin. 2019. Empowering A* search algorithms with neural networks for personalized route recommendation. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*. 539–547.

[32] Hao Wu, Jiangyun Mao, Weiwei Sun, Baihua Zheng, Hanyuan Zhang, Ziyang Chen, and Wei Wang. 2016. Probabilistic robust route recovery with spatio-temporal dynamics. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 1915–1924.

[33] Tong Xia, Yunhan Qi, Jie Feng, Fengli Xu, Funing Sun, Diansheng Guo, and Yong Li. 2021. Attnmove: History enhanced trajectory recovery via attentional network. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 4494–4502.

[34] Fudan Yu, Wenxuan Ao, Huan Yan, Guozhen Zhang, Wei Wu, and Yong Li. 2022. Spatio-Temporal Vehicle Trajectory Recovery on Road Network Based on Traffic Camera Video Data. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 4413–4421.

[35] Fudan Yu, Huan Yan, Rui Chen, Guozhen Zhang, Yu Liu, Meng Chen, and Yong Li. 2023. City-scale vehicle trajectory data from traffic camera videos. *Scientific data* 10, 1 (2023), 711.

[36] Hao Yu, Xi Guo, Xiao Luo, Weihao Bian, and Taohong Zhang. 2023. Construct Trip Graphs by Using Taxi Trajectory Data. *Data Science and Engineering* 8, 1 (2023), 1–22.

[37] Haitao Yuan, Guoliang Li, Zhifeng Bao, and Ling Feng. 2020. Effective travel time estimation: When historical trajectories over road networks matter. In *Proceedings of the 2020 acm sigmod international conference on management of data*. 2135–2149.

[38] Jing Yuan, Yu Zheng, Chengyang Zhang, Xing Xie, and Guang-Zhong Sun. 2010. An interactive-voting based map matching algorithm. In *2010 Eleventh international conference on mobile data management*. IEEE, 43–52.

[39] F Benjamin Zhan and Charles E Noon. 1998. Shortest path algorithms: an evaluation using real road networks. *Transportation science* 32, 1 (1998), 65–73.

[40] Chuanpan Zheng, Xiaoliang Fan, Cheng Wang, and Jianzhong Qi. 2020. Gman: A graph multi-attention network for traffic prediction. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 34. 1234–1241.

[41] Yu Zheng, Licia Capra, Ouri Wolfson, and Hai Yang. 2014. Urban computing: concepts, methodologies, and applications. *ACM Transactions on Intelligent Systems and Technology (TIST)* 5, 3 (2014), 1–55.

[42] Yi Zhou and Ling Shao. 2018. Aware attentive multi-view inference for vehicle re-identification. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 6489–6498.

[43] Chenyi Zhuang, Nicholas Jing Yuan, Ruihua Song, Xing Xie, and Qiang Ma. 2017. Understanding People Lifestyles: Construction of Urban Movement Knowledge Graph from GPS Trajectory.. In *Ijcai*. 3616–3623.

A TRAINING DATA PREPARATION

Algorithm 1: Training Sample Generation

Input: Shortest distance matrix D_e, current passing record r_s, next passing record r_e, possible adjacent road sections $\overline{Adj}^T_{r_s.e}$, vehicle p, the historical passing records $\mathcal{H}_p$.

Output: training samples Tr.

1 Init $pos_{dis} \leftarrow \{\}$; $neg_{dis} \leftarrow \{\}$; $\tau \leftarrow 0$;
2 **for** $e_b \in \overline{Adj}^T_{r_s.e}$ **do**
3 $dis \leftarrow D_e[r_s.e][r_e.e] - D_e[e_b][r_e.e]$;
4 **if** $dis > 0$: // far away from r_e
5 pos_{dis}.push(dis);
6 **else:** // close to r_e
7 neg_{dis}.push($|dis|$);
8 **end**
9 **if** $|pos_{dis}| > 0$ *and* $|neg_{dis}| < 0$:
10 $\tau \leftarrow 0.2$;
11 **elif** $|pos_{dis}| > 0$:
12 $\tau \leftarrow 1$;
13 **for** $e_b \in \overline{Adj}^T_{r_s.e}$ **do**
14 $dis \leftarrow D_e[r_s.e][r_e.e] - D_e[e_b][r_e.e]$;
15 **if** $dis > 0$:
16 Tr.push($< r_s.e, e_b, r_e, p, \mathcal{H}_p >$, $\frac{exp(-dis)\times\tau}{\sum_{d' \in pos_{dis}} exp(d')}$);
17 **else:**
18 Tr.push($< r_s.e, e_b, r_e, p, \mathcal{H}_p >$, $\frac{exp(-dis)\times(1-\tau)}{\sum_{d' \in neg_{dis}} exp(d')}$);
19 **end**
20 **return** Tr;

First, we directly use the popular YOLOv7 [29] to locate the coordinates of vehicles in the images captured by traffic cameras. Then, we combine such information with the lane boundaries and turn details (such as straight, left, right, U-turn, etc) defined by the installer during camera installation. Based on that, we can identify the lane of the moving vehicles from camera snapshots.

Next, given a road network G, we employ the Dijkstra algorithm to obtain the shortest distances between every pair of road segments, forming the matrix D_e, where $D_e[e_a][e_b]$ denotes the shortest distance from road segment e_a to road segment e_b. Note that, this procedure is pre-processing and does not affect model design. Based on the turning information of the vehicle's located lane and the topological structure of G, we obtain the set of possible adjacent road segments that the vehicle may head towards, denoted as $\overline{Adj}^T_{r_s.e}$.

Based on the above pre-processing, Algorithm 1 shows the training data/sample generation. The basic idea is that we categorize potential adjacent road segments into two classes based on the distance, i.e., either shorten or lengthen the distance to the destination. Specifically, it first calculates the total distance away from (or towards) the destination for each class (lines 2–7). Note that we cannot simply assume that a vehicle will necessarily head towards an adjacent road segment close to the destination, even if the probability is relatively high. Thus, if there are adjacent road segments both close to and far from the destination, we set a hyperparameter τ to 0.2 (lines 9–10). This value indicates that there is a certain likelihood that the vehicle will head towards road segments that are far from the destination. This approach enables generating diverse samples and preventing the model from consistently choosing adjacent road segments close to the destination, neglecting other potential factors such as road development levels. Finally, we use the softmax function to calculate the transition probability labels for each sample (lines 13–19).

B PARAMETER SENSITIVITY

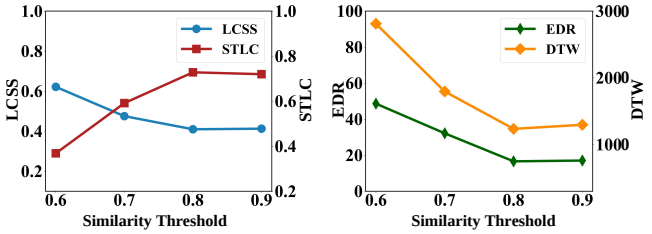


Figure 7: Different Similarity Threshold T_h.

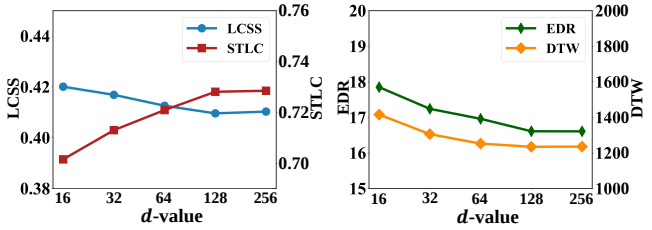


Figure 8: Different Dimension d Value.

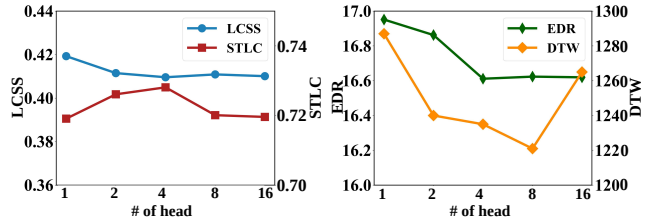


Figure 9: Different Number of Head H.

We investigate the sensitivity of three key parameters: vehicle snapshot similarity threshold T_h, the road segment representation dimension d, and the head number H, which determine the representation ability of the model. (i) The results as shown in Fig. 7 reveal that low T_h value tends to introduce more noise from other vehicles, leading to poor performance, where as a high T_h value may filter out some valid passing records. (ii) Additionally, Fig. 8 presents the performance on different values of dimension d. We observe a gradual improvement in performance as the dimension

value increases. Performance stabilizes when the dimension exceeds 128. This is because when the value of d is too small, the resulting feature vectors fail to adequately capture the information of road segments. Conversely, an excessively large dimension value introduces redundancy. Therefore, we set d to 128 to strike a balance between expressive feature representation and avoiding unnecessary redundancy. (iii) Finally, we conducted experiments using different numbers of heads H, and the results are presented in Fig. 9. We observed that when H is greater than 1, the performance is relatively stable across various values. Therefore, we chose a value of 4 as it offered a relatively better performance.